Homework Assignment #1 - Introduction, Python for CTF, and Basics of Encoding and Cryptographic Algorithms

Problem 1: Looking Under Rock

Clues:

- Famous security breach (rockyou)
- 45 being important (rockyou-45.txt)
- Puzzle looks like coordinates

```
from pathlib import Path
import re

PUZZLE_PATH = Path("/mnt/data/Puzzle")
WORDLIST_PATH = Path("/mnt/data/rockyou-45.txt")

coords = []
for line in PUZZLE_PATH.read_text(errors="replace").splitlines():
    line = line.strip()
    if not line:
        continue
    m = re.match(r"^(\d+)\s*,\s*(\d+)$", line)
    if not m:
        raise ValueError(f"Unrecognized line format: {line!r}")
    coords.append((int(m.group(1)), int(m.group(2))))

words = WORDLIST_PATH.read_text(errors="replace").splitlines()

out = []
for r, c in coords:
    if r < 0 or r >= len(words):
        raise IndexError(f"Row out of bounds: {r} (max {len(words)-1})")
    w = words[r]
    if c < 0 or c >= len(w):
        raise IndexError(f"Col out of bounds: row={r}, col={c}, word={w!r}")
    out.append(w[c])

msg = "".join(out)

print("Recovered message:")
print(msg)
print("\nFlag:")
print(f"fsuCTF{{{msg}}}")
```

Flag:

fsuCTF{YouLookedUnderSomeRocksAndFoundAWordlistThisWillBeVeryImportantLaterIReallyH

opeYouDidntDoThisManually}

Problem 2: The Mathening

```
import socket
import re
import sys

HOST = "ctf.cs.fsu.edu"
PORT = 65002

MASK64 = (1 << 64) - 1

def op_at(A, B):
    return ((~A) & (~B)) ^ B

def op_hash(A, B):
    return (((~A) | (~B)) >> 2) ^ (A << 2)

def op_merge(A, B):
    Au = A & MASK64
    Bu = B & MASK64
    out = 0
    for i in range(63, -1, -2):
        out |= Au & (1 << i)
    for i in range(62, -1, -2):
        out |= Bu & (1 << i)
    return out

OPS = {"@": op_at, "#": op_hash, "?": op_merge}

TASK_RE  = re.compile(rb"\[\d+\]\s*(-?\d+)\s*([@#\?])\s*(-?\d+)[A-Za-z]?\s*=")
WRONG_RE = re.compile(rb"Correct answer was\s+(-?\d+)")

def main():
    s = socket.create_connection((HOST, PORT))
    buf = b""

    while True:
        data = s.recv(4096)
        if not data:
            break

        sys.stdout.write(data.decode(errors="replace"))
        sys.stdout.flush()

        buf += data
```

```
        while True:
            m = TASK_RE.search(buf)
            if not m:
                break

            A = int(m.group(1))
            op = m.group(2).decode()
            B = int(m.group(3))

            ans = OPS[op](A, B)
            s.sendall(f"{ans}\n".encode())

            buf = buf[m.end():]

        m2 = WRONG_RE.search(buf)
        if m2:
            s.sendall(m2.group(1) + b"\n")
            buf = buf[m2.end():]

    s.close()

if __name__ == "__main__":
    main()
```

Flag: fsuCTF{n3w_new_n3w_m4th_tm_p4rt_2}

Problem 3: Waiter

Base64 encoding because:

- Base clue
- Labels clue indicating layering or encodings
- “JUMP ship” suggesting something that jumps meaning multiple layers
- Ends with =

Script:

```
import sys
import base64
import binascii
import re
from pathlib import Path

content = Path('note3.txt').read_bytes().strip()
layers = 0
```

```
while True:
	try:
		data = base64.b64decode(content, validate=True)
	except binascii.Error:
		break
	try:
	txt = data.decode('ascii')
	if re.fullmatch(r"[A-Za-z0-9+/=\n\r]+", txt) and len(txt) > 100:
		layers += 1
		content = txt.encode('ascii')
		continue
	else:
		layers += 1
		last_binary = data
		break
	except UnicodeDecodeError:
		layers += 1
		last_binary = data
		break

try:
	final2 = base64.b64decode(last_binary, validate=True)
	try:
		final_plain = base64.b64decode(final2, validate=True)
		print(final_plain.decode('utf-8'))
	except binascii.Error:
		print(final2[:256].hex())
	except UnicodeDecodeError:
		print(final_plain[:256].hex())
except binascii.Error:
	try:
		print(last_binary.decode('utf-8'))
	except UnicodeDecodeError:
		print(last_binary[:256].hex())
```

Flag: fsuCTF{T0_0ff3nd_4_Co0k_47_S34_Is_4_F001s_Mist4k3}

Problem 4: Endangered Program

I opened it up in notepad first and saw this line, “VIRTUAL_ENV The endangered program loves this environment.” After that I created a Python virtual environment in WSL file system through powershell doing the following:

Python3 -m venv gift

source ~/gift/bin/activate

Then I ran the program while the venv was active.

Flag: fsuCTF{7hi5_I5_7h3_P3rf3c7_3nvir0nm3n7}